

QUESTIONS

LEVEL 2 — MEDIUM

Q1

function



Q2

1DArray



Q3

string



Q4

2DArray



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Level 2 — Medium factorial (M) — function

✓ Submitted

Write a function `int factorial(int n)` that calculates the factorial of a given number `n`.

Sample Input:

Enter a number: 5

Sample Output:

Factorial of 5 is 120

Your Code

```
1 #include<iostream>
2 using namespace std;
3 int main() {
4     int n,fact=1;
5     cout<<"Enter a Number: ";
6     cin>>n;
7     for(int i=1;i<=n;i++)
8     {
9         fact=fact*i;
10    }
11    cout<<"\nFactorial: "<<fact;
12    return 0;
13 }
```

Output

```
Enter a Number:
Factorial: 120
```

LEVEL 2 — MEDIUM

Q1

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2DArray



4 / 4 saved

Write a program to check if a matrix is symmetric.

Sample Input:

Enter number of rows: 3

Enter number of columns: 3

Enter elements:

1 2 3

2 4 5

3 5 6

Sample Output:

The matrix is symmetric.

Test Case 1: Enter number of rows: 3 Enter number of columns: 3 Enter elements: 1 2 3 4 5 6 7 8 9

Test Case 2: Enter number of rows: 2 Enter number of columns: 2 Enter elements: 1 2 3 4

Test Case 3: Enter number of rows: 4 Enter number of columns: 4 Enter elements: 1 0 0 0 0 2 0 0 0 0 3 0 0 0 0 4

Your Code

```
1 #include<iostream>
2 using namespace std;
3 int main() {
4     int r, c, a[10][10];
5     bool symmetric = true;
6     cout<<"Enter number of rows: ";
7     cin>>r;
8     cout<<"\nEnter number of columns: ";
9     cin>>c;
10    cout<<"\nEnter elements:";
```

Output

```
Enter number of rows:
Enter number of columns:
Enter elements:
The matrix is symmetric
```

Q1

function



Q2

1DArray



Q3

string



Q4

2DArray



4 / 4 saved

Write a program to check if a given string is a palindrome.

Sample Input:

Enter a string: madam

Sample Output:

The string is a palindrome.

Test Case 1: Enter a string: Hello

Test Case 2: Enter a string: Python

Test Case 3: Enter a string: Programming

Your Code

```
1 #include<iostream>
2 #include<string>
3 using namespace std;
4 int main() {
5     string str, rev = "";
6     cout<<"Enter a String: ";
7     cin>>str;
8     for(int i = str.length()-1;i>=0;i--)
9     {
10         rev = rev+str[i];
11     }
12     if(str==rev)
13         cout<<"\nThe string is a palindrome";
14     else
15         cout<<"\nThe string is not a palindrome";
16     return 0;
```

Output

```
Enter a String:
The string is a palindrome
```

Q1
function



Q2
1DArray



Q3
string



Q4
2DArray



4 / 4 saved

Write a program to find the missing number in an array of integers from 1 to n.

Sample Input:

Enter number of elements: 4

Enter elements: 1 2 4 5

Sample Output:

Missing number: 3

Test case1:

Enter number of elements: 5

Enter elements: 1 2 3 5 6

Test case2:

Enter number of elements: 7

Enter elements: 1 2 3 4 5 6 8

Test case3:

Enter number of elements: 6

Enter elements: 1 3 4 5 6 7

Your Code

```
1 #include<iostream>
2 using namespace std;
3 int main() {
4     int n;
5     cout<<"Enter number of elements: ";
6     cin>>n;
7     int a[100];
8     cout<<"\nEnter elements: ";
9     int sum = 0;
10    for(int i=0; i<n; i++) {
11        cin>>a[i];
12        sum = sum + a[i];
```

Output

```
Enter number of elements:
Enter elements:
Missing Number: 3
```